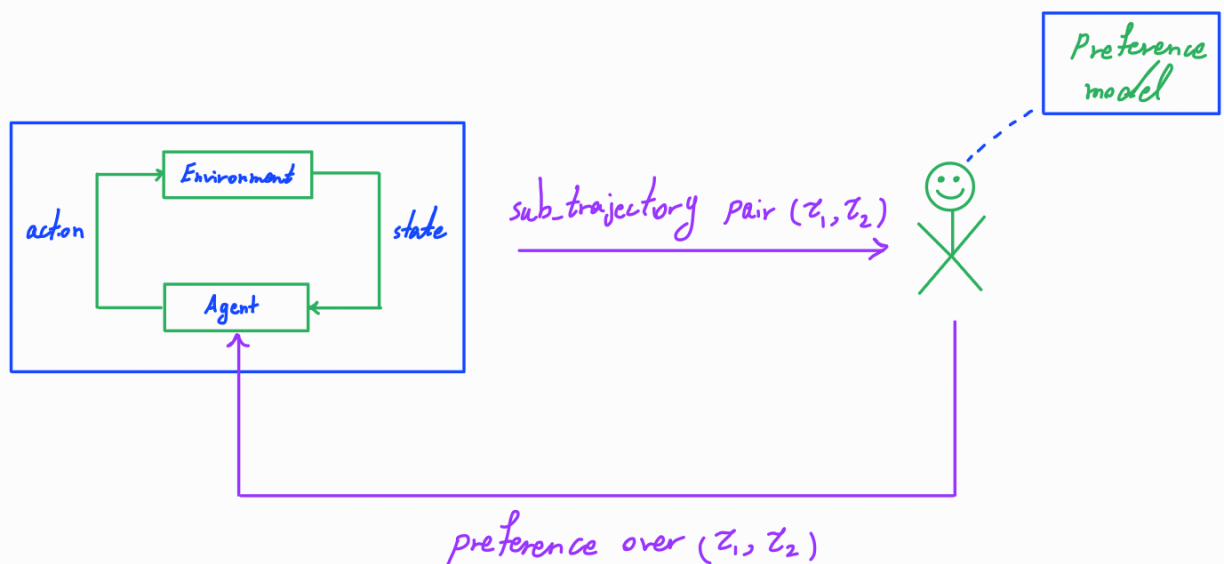


Preference-Based Learning

There are settings where, similar to imitation learning,

- a natural reward function does not exist,
- it is difficult to form a reward function, or
- it is difficult to learn from a reward function.

Additionally, expert demonstrations are unavailable or difficult to obtain; however, high-quality data in the form of preferences (comparative feedback) over different behaviors is available.



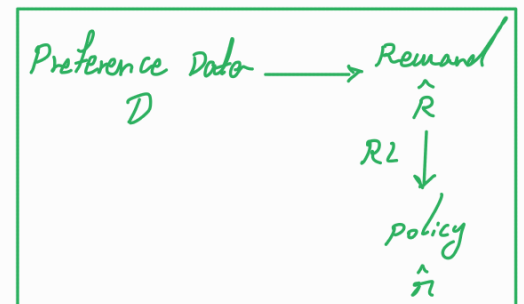
In these settings, preference-based learning provides a framework to learn a (near-) optimal policy from the preferences (comparative feedback).

Beyond RL, preference-based learning has also become adopted for fine-tuning and alignment of other machine learning models, e.g., language models.

Benefits

- No need for the human to be expert, just capable of judging.
- Fewer queries from human compared to querying reward.
- Alignment with human preferences, allows personalization.

Reinforcement learning from human feedback (RLHF)



Direct preference optimization (DPO)

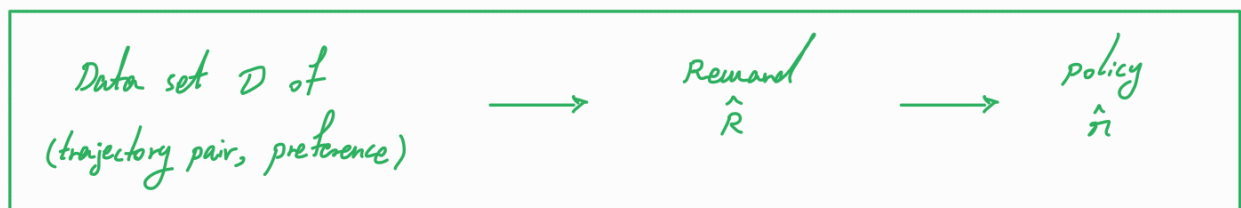


Reinforcement Learning from Human Feedback (RLHF)

RLHF is a reinforcement learning paradigm where the agent's feedback is given by a human, often in the form of ranking instead of rewards.

Idea:

- Find the underlying reward function $\hat{R}$ by maximizing the likelihood of the data set of (trajectory pair, preference).
- Then, apply RL algorithms or their modified versions over the learned reward function $\hat{R}$.



* To this end, we need to assume the existence of a preference model and measure the likelihood of preference data with respect to that model.

Preference Model

Suppose two sub-trajectories z_1 and z_2 are observed by human.

We assume that the human's probability of preferring z_1 over z_2

follows the Bradley-Terry model:

Plackett-Luce is a generalization of this

Bradley-Terry Model

items / decisions x_i, x_j

utility function $u: \mathcal{X} \rightarrow \mathbb{R}_+$

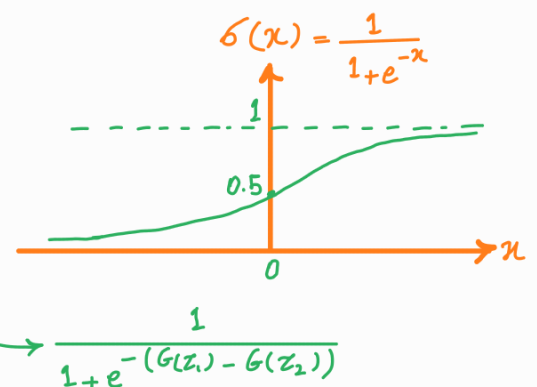
preferred over

$$P[x_i \succ x_j] = \frac{u(x_i)}{u(x_i) + u(x_j)}, \quad P[x_j \succ x_i] = \frac{u(x_j)}{u(x_j) + u(x_i)},$$

where $(x_i \succ x_j) = (x_i < x_j)^c = (x_j \succ x_i)^c$.

$$P[z_1 \succ z_2] = \frac{\exp(G(z_1))}{\exp(G(z_1)) + \exp(G(z_2))}$$

$$= \sigma(G(z_1) - G(z_2))$$



$$P[z_2 \succ z_1] = 1 - P[z_1 \succ z_2] = \sigma(G(z_2) - G(z_1))$$

where

$$G(z = (s_0, a_0, s_1, a_1, \dots, s_l, a_l)) = \sum_{t=0}^l \gamma^t r(s_t, a_t).$$

Maximum Likelihood Objective: Following the probabilistic preference model, the likelihood of data point (z_1, z_2, μ) is

labeled data $\rightarrow$ supervised learning
trajectory pair $\rightarrow$ preference label

$$P[(z_1, z_2, \mu)] = \begin{cases} P[z_1 \succ z_2] = \sigma(G(z_1) - G(z_2)) & \text{if } \mu = (z_1 \succ z_2) \\ P[z_2 \succ z_1] = \sigma(G(z_2) - G(z_1)) & \text{if } \mu = (z_2 \succ z_1) \end{cases}$$

For a data set of preferences $\mathcal{D} = \{(z_1^i, z_2^i, \mu^i)\}_{i=1}^N$, assuming the data points are independent conditioned on the reward,

$$P((z_1^i, z_2^i, \mu^i)_{i=1}^N | r) = \prod_{i=1}^N P((z_1^i, z_2^i, \mu^i) | r)$$

$$P[\mathcal{D}] = \prod_{i=1}^N \left(\mathbb{1}[z_1^i \succ z_2^i] \frac{P[z_1 \succ z_2]}{\sigma(G(z_1) - G(z_2))} + \mathbb{1}[z_2^i \succ z_1^i] \frac{P[z_2 \succ z_1]}{\sigma(G(z_2) - G(z_1))} \right)$$

and the maximum likelihood estimation (MLE) loss function

$$L(r) = -\log P[\mathcal{D}]$$

can be written as

$$\text{human preference } \mu(1) \geq 0, \mu(2) \geq 0, \mu(1) + \mu(2) = 1$$

$$L(r) = -\sum_{i=1}^N (\mu(1) \log \sigma(G(z_1) - G(z_2)) + \mu(2) \log \sigma(G(z_2) - G(z_1)))$$

and optimized to obtain $\hat{r}$:

$$\hat{r}_{\hat{\theta}} = \arg \min_{r_{\theta}} L(r_{\theta})$$

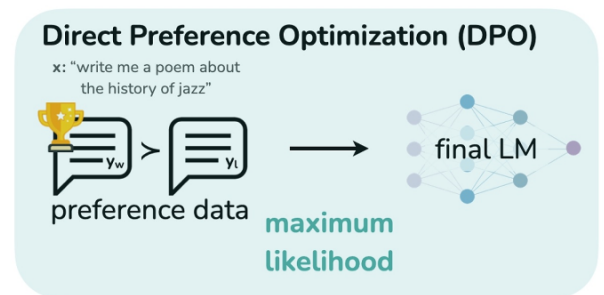
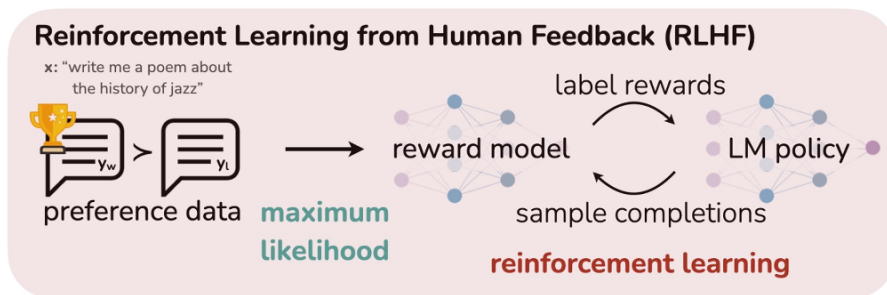
e.g., using SGD

Notes [Variations and Extensions of RLHF]: There are many variations and extensions of RLHF, such as

- Not using discounting in the preference model
- Handling comparison of trajectory pairs with different starting state
- Using regret instead of return in the preference model
- Sampling segments from trajectories in a replay buffer
- Extension to k -wise comparison (ranking) $\rightarrow$ Plackett-Luce model
- Allowing abstention voting
- Active querying
- Extension to non-Markovian rewards

Direct Preference Optimization (DPO)

Learning a reward function in RLHF could be a complex and unstable procedure. Instead of that, DPO directly learns an optimal policy.



DPO for Fine-Tuning: Consider a language model that has been trained, behaving according to a policy $\pi_{\text{ref}} : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$.

This model can be queried to generate a preference query

$$a_1 \sim \pi_{\text{ref}}(\cdot | s) \quad , \quad a_2 \sim \pi_{\text{ref}}(\cdot | s)$$

and a human label u

$$a_1 \succ a_2 \quad \text{or} \quad a_2 \succ a_1 .$$

The Bradley-Terry model assumes that the preference is modeled by

$$P[a_1 \succ a_2 | s] = \frac{\exp(r^*(s, a_1))}{\exp(r^*(s, a_1)) + \exp(r^*(s, a_2))}$$

for a true reward function $r^*(\cdot, \cdot)$.

Maximum Likelihood Objective: Given a data set of preferences

$\mathcal{D} = \{(s^i, a_1^i, a_2^i, \mu^i)\}_{i=1}^N$, the MLE objective to find θ^* corresponding to r^* is

$$L(r_\theta) = - \mathbb{E}_{(s, a_1, a_2, \mu) \sim \mathcal{D}} \left[\underbrace{\mathbb{1}[a_1 \succ a_2]}_{\mu(1)} \log \sigma(r_\theta(s, a_1) - r_\theta(s, a_2)) + \underbrace{\mathbb{1}[a_2 \succ a_1]}_{\mu(2)} \log \sigma(r_\theta(s, a_2) - r_\theta(s, a_1)) \right]$$

By minimizing $L(r_\theta)$, we can obtain $\hat{\theta}$.

RL Fine-Tuning: The language model can then be fine-tuned using RL over the learned reward function

$$\max_{\pi_\theta} \underbrace{\mathbb{E}_{\substack{s \sim \mathcal{D} \\ a \sim \pi_\theta(\cdot|s)}} [r_{\hat{\theta}}(s, a)]}_{\text{return}} - \beta \underbrace{\mathbb{E}_{s \sim \mathcal{D}} [D_{KL}(\pi_\theta \| \pi_{\text{ref}})]}_{\text{deviation from the reference policy}}$$

DPO Objective: It can be shown that the above optimization problem has a closed-form solution for $\pi_{\hat{\theta}}$ in terms of $r_{\hat{\theta}}$. Vice versa, the reward $r_{\hat{\theta}}$ can be written in terms of $\pi_{\hat{\theta}}$.

$$\pi_{\hat{\theta}} = f(r_{\hat{\theta}}) \longrightarrow r_{\hat{\theta}} = f^{-1}(\pi_{\hat{\theta}})$$

* Doing so, we can write the Bradley-Terry preference model in terms of $\pi_{\hat{\theta}}$ or π_{θ^*} instead of r^* :

$$P[a_1 \succ a_2 | s] = \frac{1}{1 + \exp\left(\beta \log \frac{\pi_{\theta^*}(a_2 | s)}{\pi_{\text{ref}}(a_2 | s)} - \beta \log \frac{\pi_{\theta^*}(a_1 | s)}{\pi_{\text{ref}}(a_1 | s)}\right)}.$$

* Therefore, the MLE objective can also be written in term of a parameterized policy π_{θ} :

$$L(\pi_{\theta}) = -\mathbb{E}_{(s, a_1, a_2, \mu) \sim \mathcal{D}} \left[\mathbb{1}[a_1 \succ a_2] \log \sigma\left(\beta \log \frac{\pi_{\theta}(a_2 | s)}{\pi_{\text{ref}}(a_2 | s)} - \beta \log \frac{\pi_{\theta}(a_1 | s)}{\pi_{\text{ref}}(a_1 | s)}\right) + \mathbb{1}[a_2 \succ a_1] \log \sigma\left(\beta \log \frac{\pi_{\theta}(a_1 | s)}{\pi_{\text{ref}}(a_1 | s)} - \beta \log \frac{\pi_{\theta}(a_2 | s)}{\pi_{\text{ref}}(a_2 | s)}\right) \right]$$

and optimized to obtain $\pi_{\hat{\theta}}$:

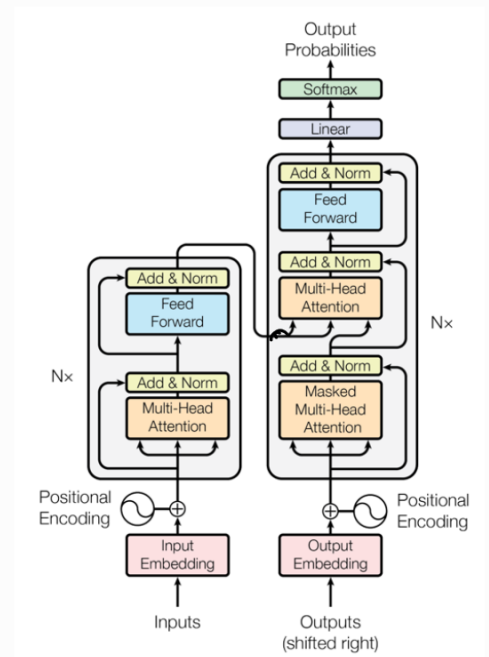
e.g., using SGD

$$\pi_{\hat{\theta}} = \arg \min_{\pi_{\theta}} L(\pi_{\theta}).$$

Case Study : Chat GPT

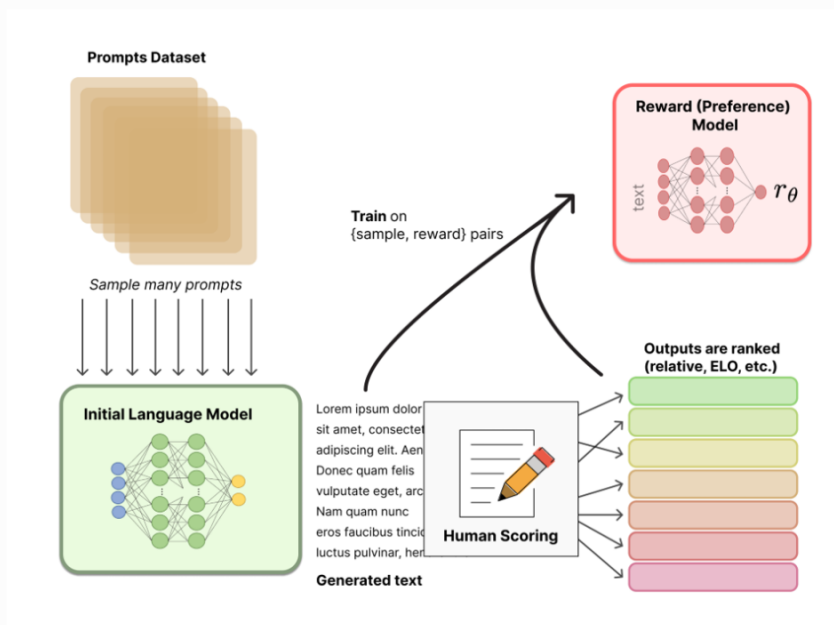
Large language models nowadays heavily rely on Transformer models.

- word/vec embedding
 - positional embedding
 - contextual embedding
- based on attention mechanisms



Encoder-Decoder design

RLHF for fine-tuning LLM: Human comparative feedback can be used to fine-tune a trained model through RLHF.



RL Feedback

